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➤ IMPLEMENTATION OF ROBOT GRIPPER IN
AUTODESK FUSION 360

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'Formulating an Adaptive Robot Gripper for Improved Object Manipulation'

The goal of this project is to use Fusion 360 to design and create an adaptive robot gripper that overcomes the constraints of traditional grippers in terms of variety and grasping ability. Automation operations are less efficient since the current robot grippers frequently struggle to grasp items of various sizes and forms. Robotics has a huge problem in developing a gripper that can adapt to various objects while keeping a firm hold.

Introduction:

Robot grippers are crucial parts of robotic systems that are intended to grab, handle, and manipulate items. Robot grippers enable robots to carry out comparable duties in a variety of industries, including manufacturing, logistics, healthcare, and research. Our hands allow humans to interact with and pick up items.

A robot gripper's main job is to firmly grab and hold items of various sizes, shapes, and materials. They are available in a broad variety of designs, each one tailored for a particular need.

To provide a firm hold and prevent hurting the things they touch, modern robot grippers frequently have sensors and feedback devices. For precise and controlled manipulation, they can be mounted on robotic hands, robotic arms, or other robotic end-effectors.

The choice of an appropriate gripper relies on the particular application since elements like item size, weight, form, and material are very important. As robotics and material science develop, gripper designs get better, becoming more adaptable and able to handle a wider range of jobs.

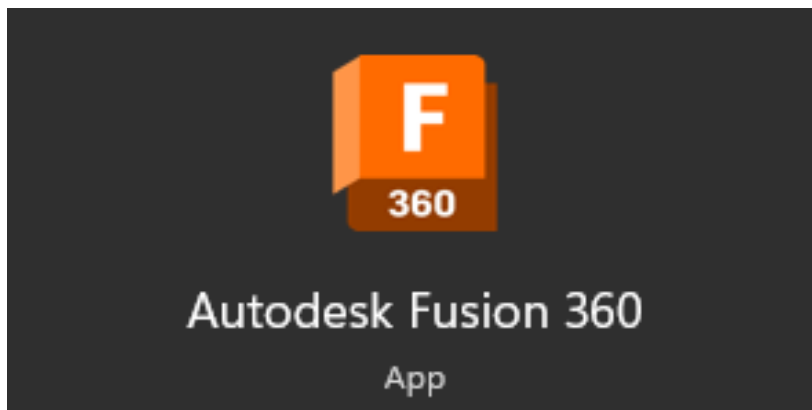
Due of its versatility in handling a variety of objects and materials, robot grippers are used extensively across a wide range of sectors. Here are a few typical uses for robot grippers:

- **Manufacturing and assembly:** Grippers are frequently used to pick up and move pieces throughout the production process on manufacturing and assembly lines. They make ensuring that components are handled consistently and accurately, accelerating assembly and minimizing errors.
- **Logistics & Warehousing:** Robot grippers assist in picking, packing, and sorting activities in warehouses and distribution centers. They handle various boxes, parcels, and commodities for shipping with efficiency.
- **Material Handling:** Grippers are used to load and unload items into conveyors, vehicles, or pallets in material handling systems. They do this by automating heavy lifting activities, which increases productivity and safety.
- **Electronics assembly:** Care must be used when handling delicate electronic components. To enable delicate and precise positioning of these components on circuit boards, grippers with sensor feedback are used.
- **Food Processing:** To handle and package food products in the food business, grippers made of food-grade materials are utilized, providing hygienic and effective handling.
- Robot grippers are used in the medical field to help with patient care chores, surgical operations, handling and positioning of surgical equipment, and other jobs.
- **Agriculture:** The automated harvesting, sorting, and packing of fruits and vegetables are among automated agricultural applications that use grippers.
- **Research and development:** In their lab experiments, researchers explore robotics, AI, and automation theories using sophisticated grippers. A new robotic system's development and testing heavily rely on these grippers.
- **Garbage Management:** To properly handle hazardous garbage and sift recyclable products, robotic grippers are utilized in waste management systems.

- **Automotive Industry:** In the production of automobiles, grippers are used for jobs like handling and assembling components.
- Robot grippers are used in the aerospace industry to help assemble aero plane parts and complete maintenance jobs.

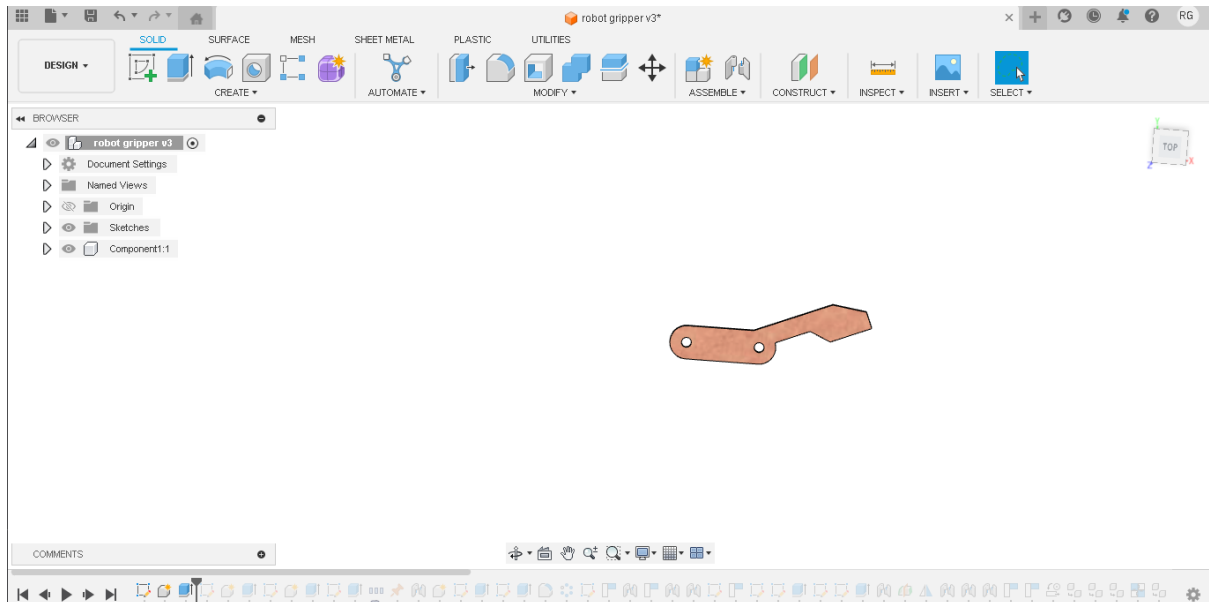
Robot gripper applications are anticipated to increase as robotics technology develops, spreading into new fields and enabling automation across numerous sectors.

Software Used:

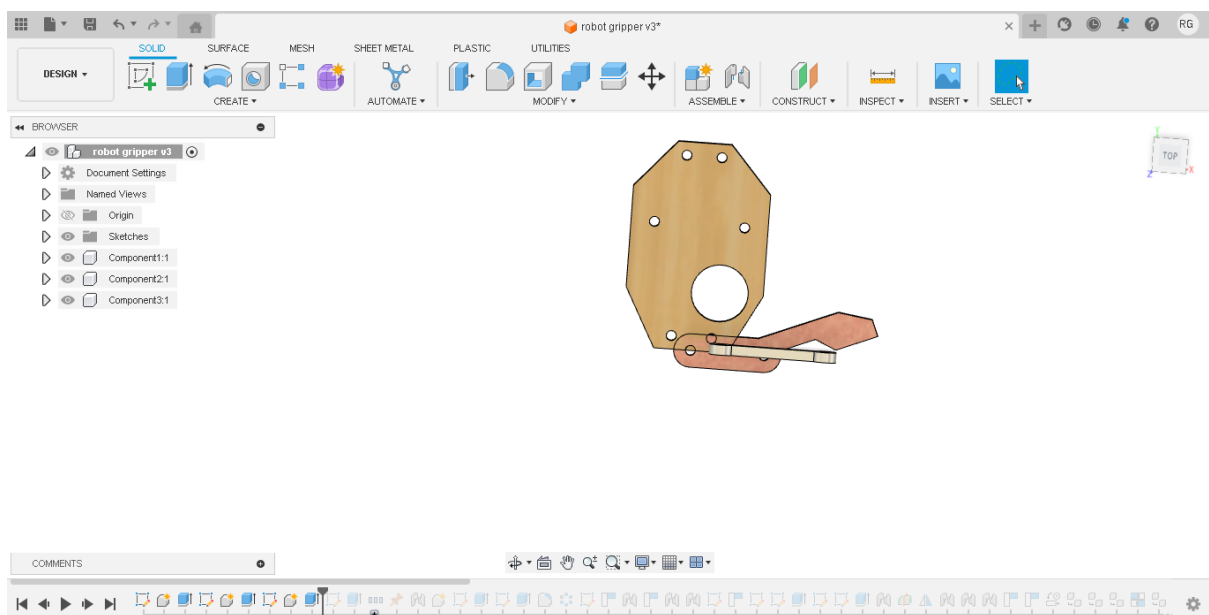


Implementation :

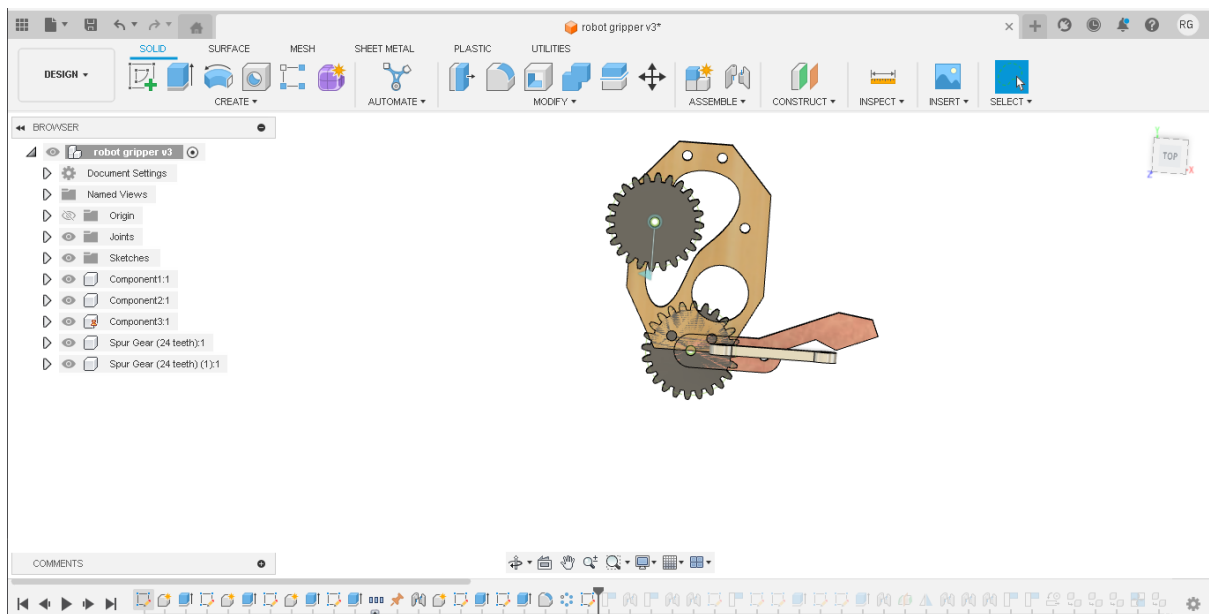
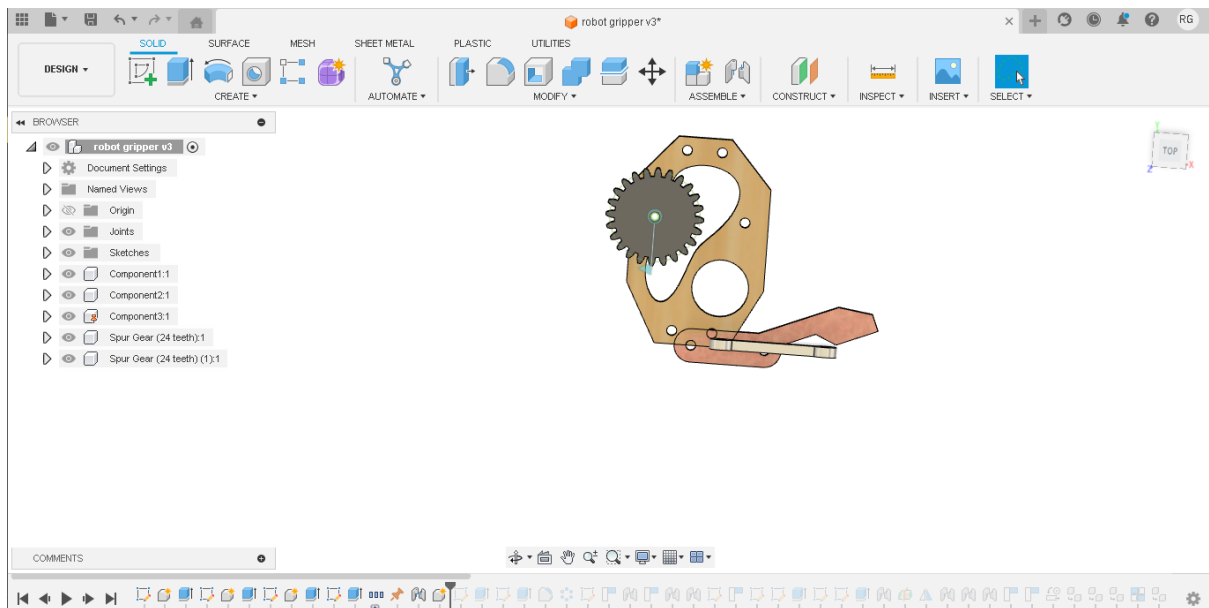
1. Making of an arm

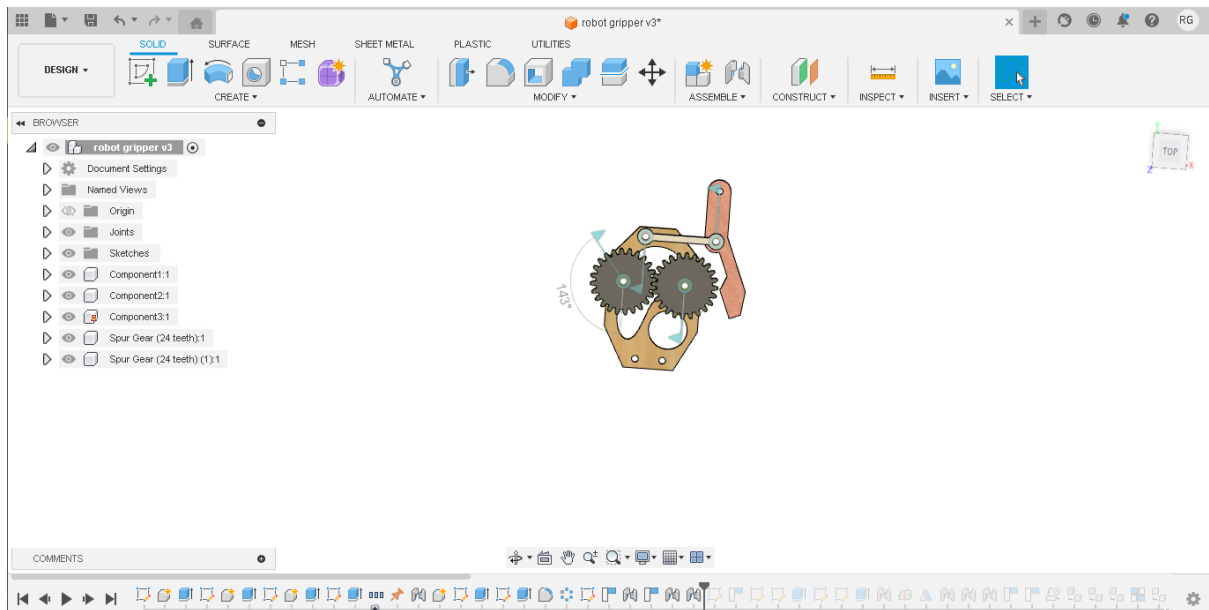


2. Making of base, arm and the support

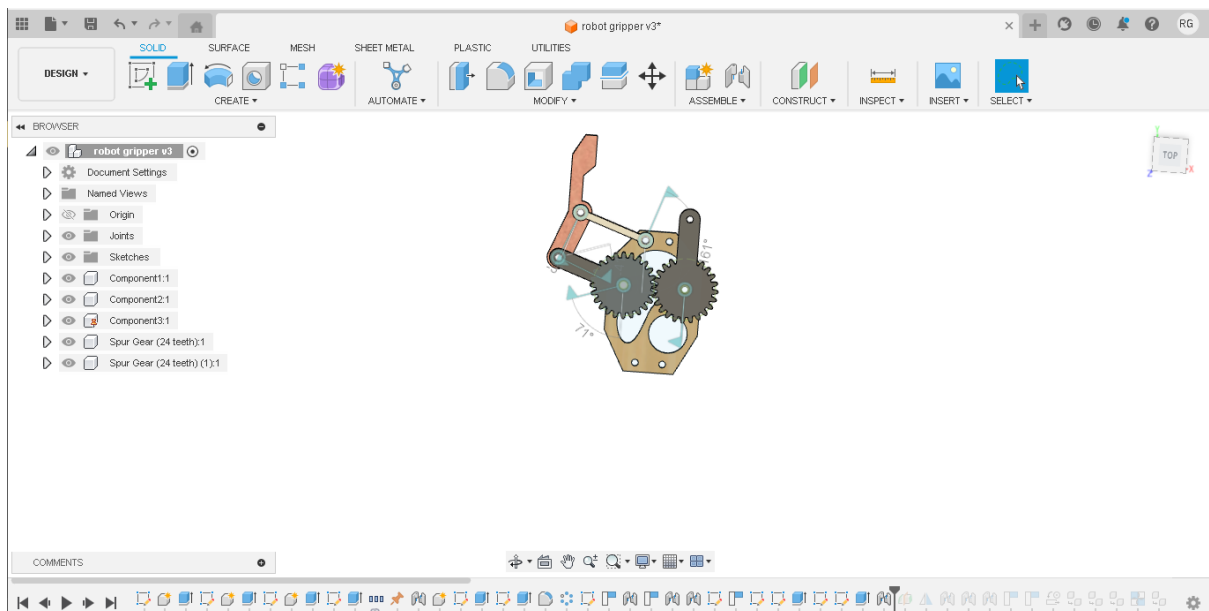


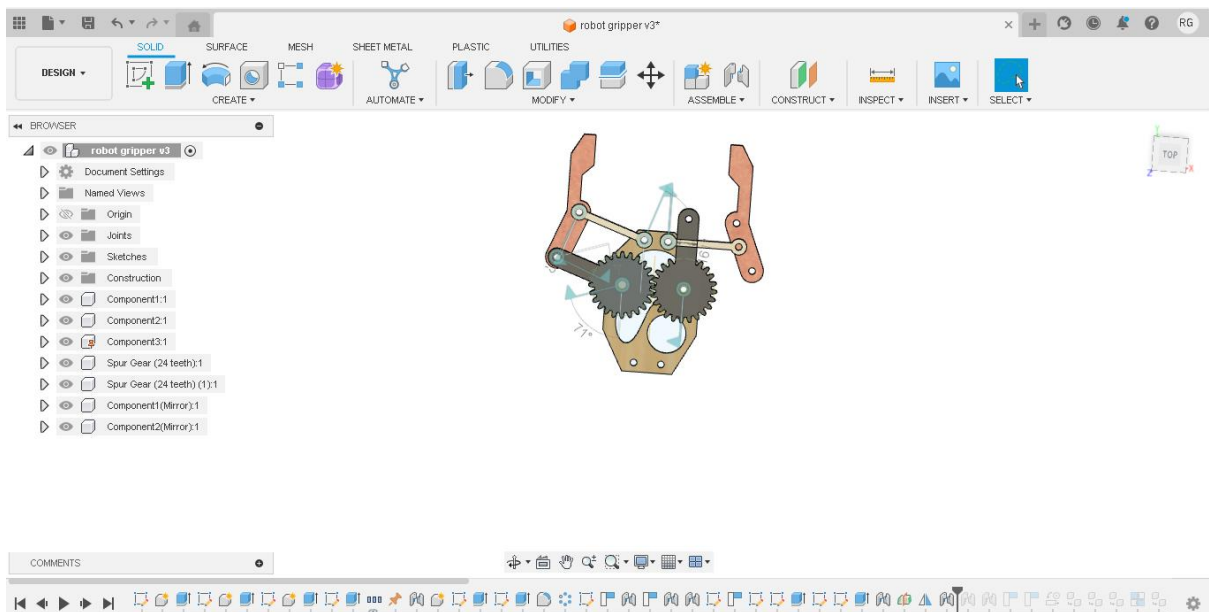
3. Fixing the spur gear



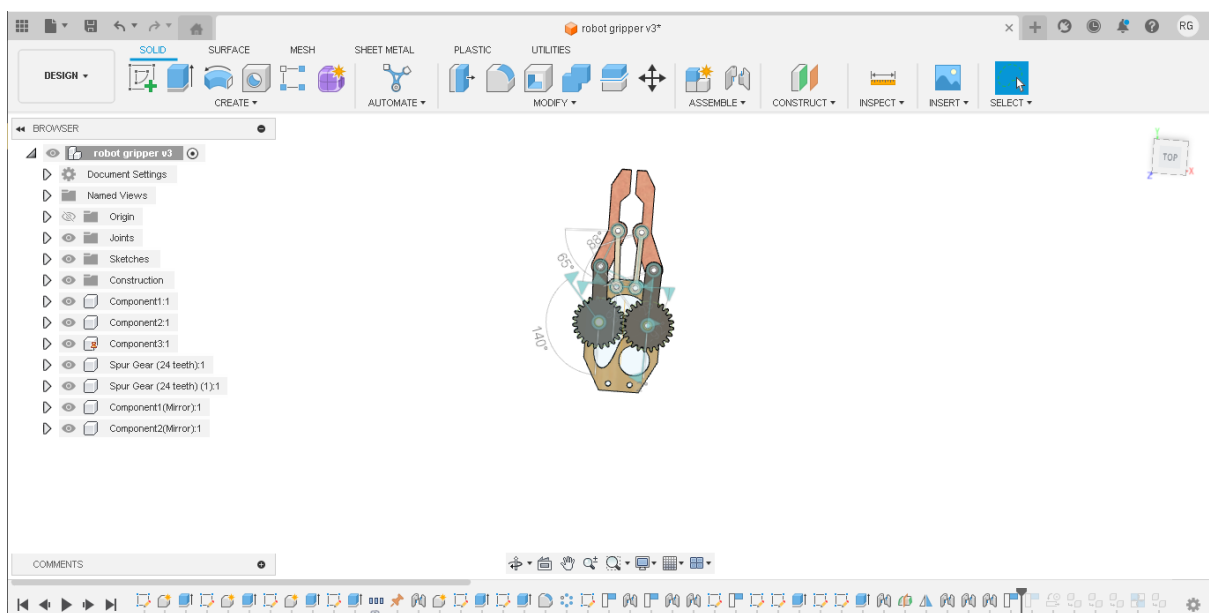


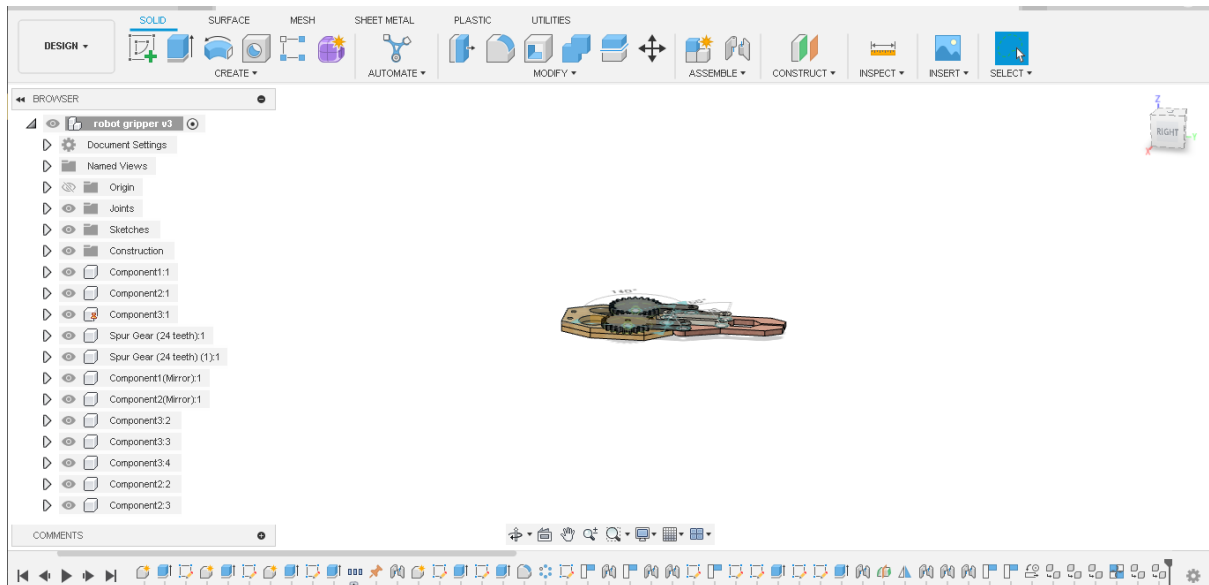
4. Mirroring of components



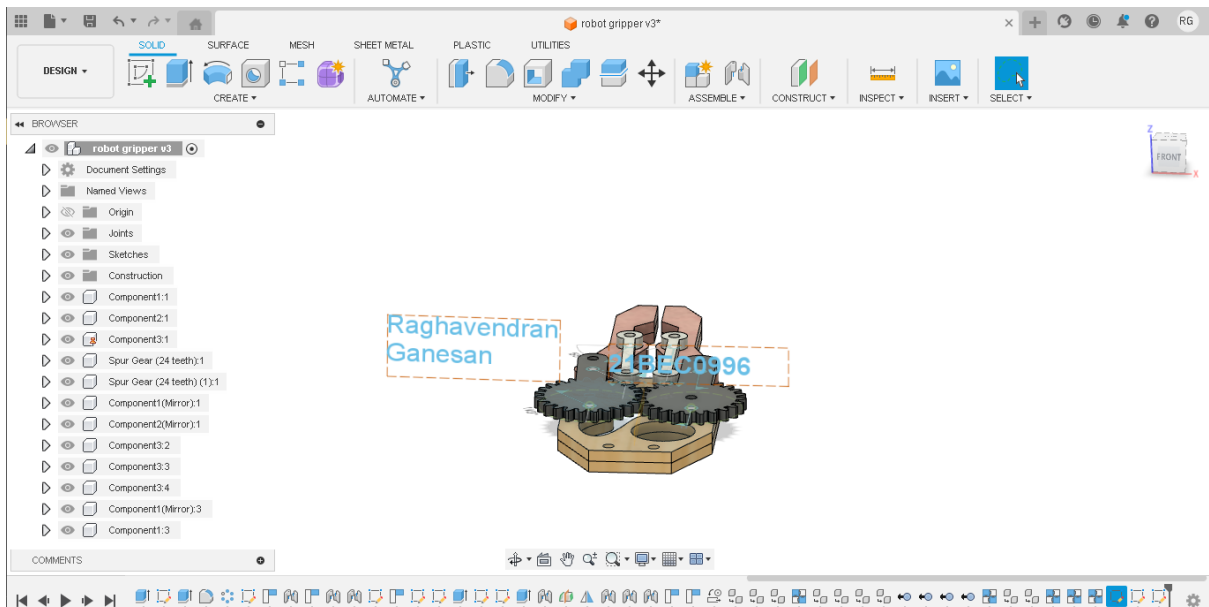


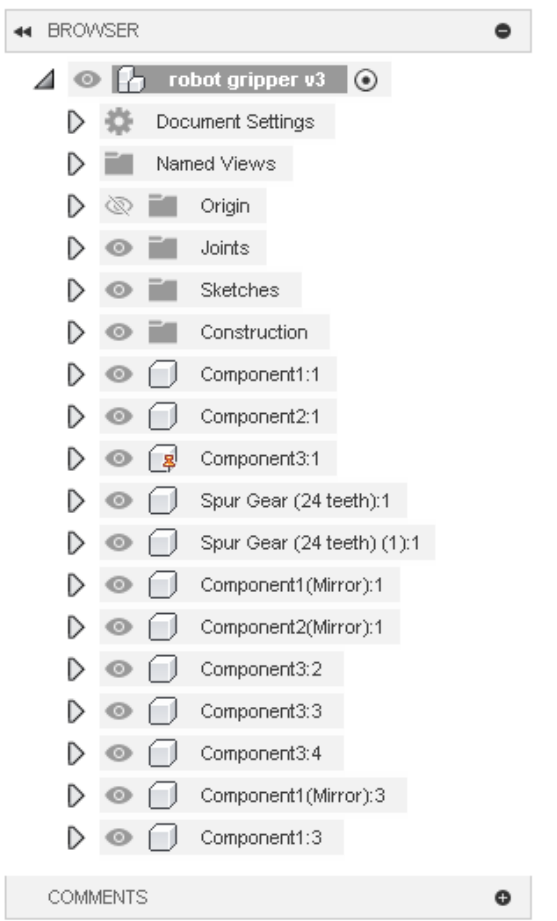
5. Making the components Rigid



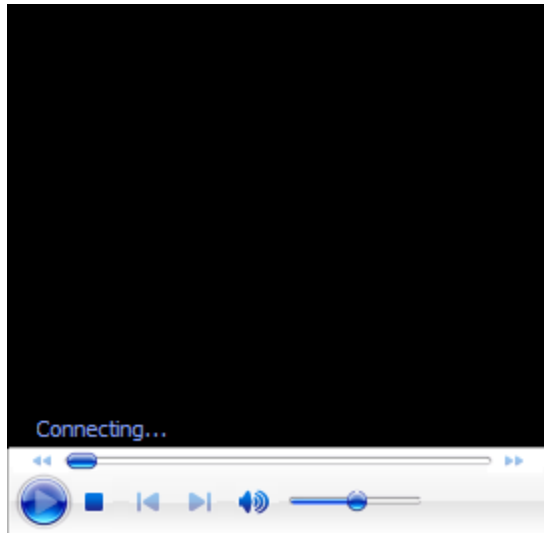


6. Final Schematic



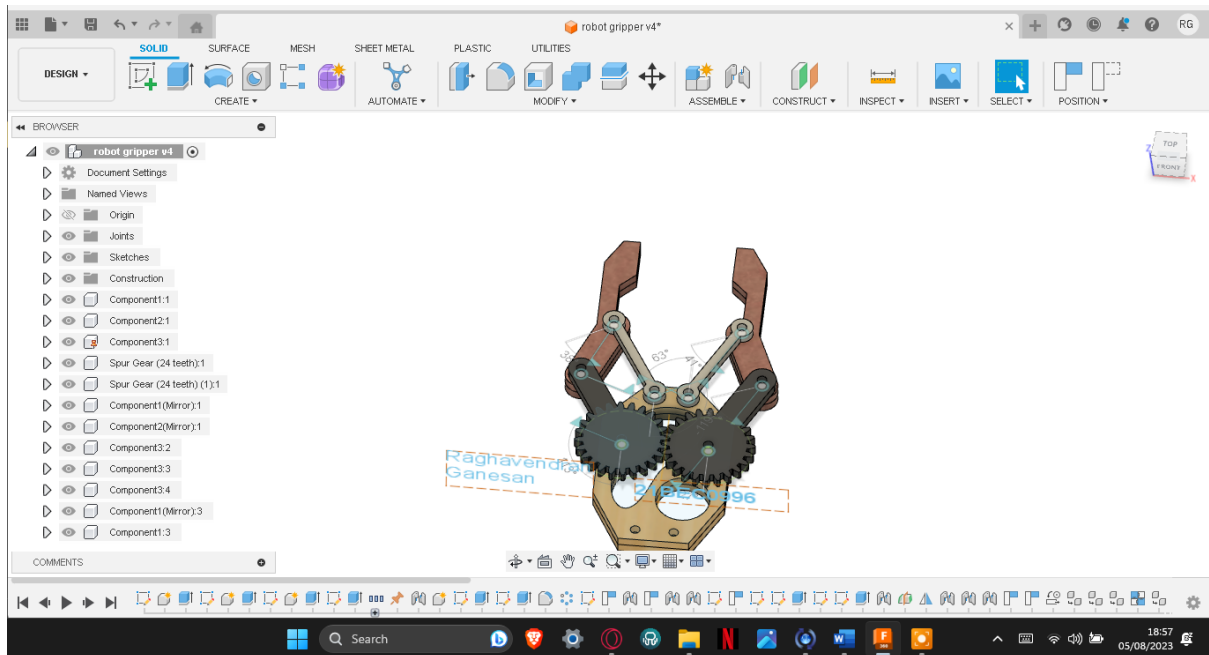


Proof of Concept :



Google drive link :

https://drive.google.com/file/d/1JOS_m7IT7y5xv-7OokCm2zHDSJfEfWBy/view?usp=sharing



References :

1. Spark Plug : <https://www.youtube.com/watch?v=ZmckF8zYbp0>
2. SolidWorks : <https://www.youtube.com/watch?v=Z4vRkZ8kcTU>
3. Fusion 360 : <https://www.youtube.com/watch?v=Z4vRkZ8kcTU>